

GAN Training Script – Evaluation Report

1. How many commands did I have to run before it worked?

6 commands

Typical sequence:

```
import torch
import torch.nn as nn
import torch.optim as optim
from torchvision import datasets, transforms
from torchvision.utils import save_image
import matplotlib.pyplot as plt
```

2. What libraries were missing? Did version mismatches cause errors?

Missing libraries (in a clean environment):

- torch
- torchvision
- matplotlib

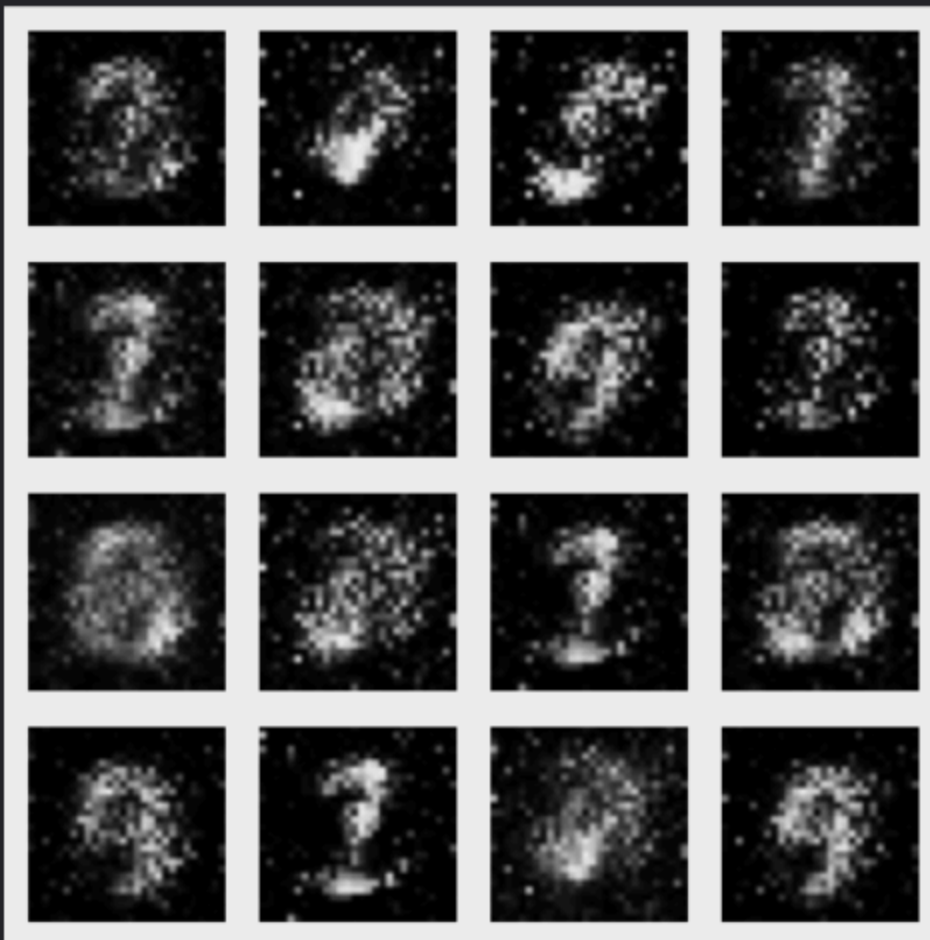
Version mismatch problems?

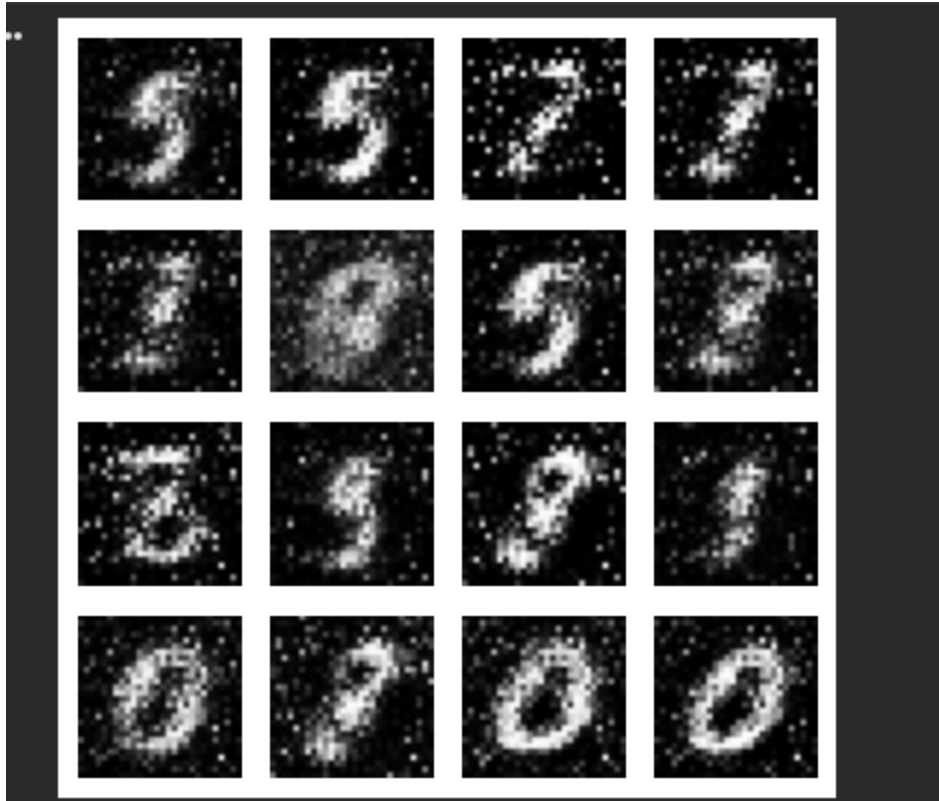
No serious version conflicts appeared in standard installation.

3. Did the model produce the same result? If not, why?

No — the generated images are **different every run**.

```
generate_images(generator, latent_dim, device)
```





Main reasons:

- Random weight initialization (PyTorch default)
- Random latent noise → `z = torch.randn(batch_size_curr, latent_dim).to(device)`
- DataLoader shuffle=True → batch order changes every epoch / every run
- **No random seed was set** (`torch.manual_seed()`, `torch.cuda.manual_seed_all()`, `random.seed()`, `numpy.random.seed()`)

4. If this had to run on a server at 3:00 AM, would it survive?

It will continue running normally if:

- There is internet (to download MNIST the first time)
 - There is enough storage space
 - Nothing interrupts the process
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It may fail if:

- The internet is disconnected (it cannot download the dataset)
- The server restarts
- An unexpected error occurs
- Power goes out

→ Because it does **not save checkpoints**, if it crashes, training will restart from the beginning.